Foundations of Programming Lab Assignments 16 – 22

A#16 – Even/Odd A#17 – Sine Series A#18 – Strings A#19 – Employee A#20 – Point

ASSIGNMENT NO. 22

Menu-Driven String Operations (Built-in Functions)

do-while menu using strlen, strcpy, strcat, strcmp, and manual reversal. Loop repeats until user chooses Exit.

C CODE

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s1[100], s2[100];
    int  choice, i, len;

    printf("Enter first string : "); scanf("%s", s1);
    printf("Enter second string: "); scanf("%s", s2);

    do {
        printf("\n═══ String Operations Menu ═══\n");
        printf("1. Length      (strlen)\n");
        printf("2. Copy        (strcpy)\n");
        printf("3. Concatenate (strcat)\n");
        printf("4. Compare     (strcmp)\n");
        printf("5. Reverse s1\n");
        printf("6. Exit\n");
        printf("Choice: "); scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Length = %d\n", (int)strlen(s1)); break;
            case 2: {
                char copy[100]; strcpy(copy, s1);
                printf("Copied: %s\n", copy); break; }
            case 3: {
                char cat[200]; strcpy(cat, s1); strcat(cat, s2);
                printf("Concatenated: %s\n", cat); break; }
            case 4:
                printf(strcmp(s1, s2) == 0 ? "EQUAL\n" : "NOT equal\n"); break;
```

```
            case 5: {
                char rev[100]; len = strlen(s1);
                for (i = 0; i < len; i++) rev[i] = s1[len - 1 - i];
                rev[len] = '\0';
                printf("Reversed: %s\n", rev); break; }
            case 6:
                printf("Exiting...\n"); break;
            default:
                printf("Invalid choice!\n");
        }
    } while (choice != 6);

    return 0;
}
```

SAMPLE I/O

```
Enter first string : Hello
Enter second string: World

Choice 3 → Concatenated: HelloWorld
Choice 1 → Length = 5
Choice 5 → Reversed: olleH
Choice 4 → NOT equal
Choice 6 → Exiting...
```

BUILT-IN FUNCTIONS

STRLEN(S)
Length — chars before '\0'

STRCPY(D,S)
Copy string s into d

STRCAT(D,S)
Append s to end of d

STRCMP(A,B)
0=equal, else non-zero

FLOWCHART

Syntax error in text
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